Hydrogen and helium abundance histories in non-tilted axisymmetric Bianchi I cosmology

Working research manuscript

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Abstract

This workspace describes a homogeneous hydrogen–helium reionization calculation and its matched signed-shear comparisons. The principal products are nuclear charge-state abundances versus elapsed gas proper time since the selected initial slice and time-ordered gas temperature. Numerical histories and figures will be inserted only from completed, identified calculations. This document includes actual short coupled pilot histories over mean redshift 20 to 19.8, with scoped independent pilot admission. The full mean-redshift 20 to 4 campaign, empirical refinement and resolved paired-response results remain pending.

1 Geometry, clocks, and controlled approximation

The metric is

$$ds^2 = -c^2dt^2 + a^2\{e^{-2b}(dx^2 + dy^2) + e^{4b}dz^2\}. \tag{1}$$

With $s = \dot{b}$, the directional expansion rates are $(H - s, H - s, H + 2s)$ and $\sigma^2 = 3s^2$. The primary physical family keeps matter, radiation and Λ densities fixed and uses $H^2 = H^2_{\mathrm{FLRW}} + s^2$, $s \propto a^{-3}$. A separate SAME_H calculation is a counterfactual diagnostic. The reference label $q_{\mathrm{ref}} = s_*/H_{\mathrm{FLRW}}(a_*)$ differs from the instantaneous $\Sigma = s/H$. The minimum family is $q_{\mathrm{ref}} = 0, \pm q_{\mathrm{amp}}/2, \pm q_{\mathrm{amp}}$; amplitude and reference epoch require an identified scenario and quantified omitted-feedback bound.

The gas is non-tilted. In the coordinate convention $x^0 = ct$, its unit normal satisfies $n \cdot n = -1$ and physical four-velocity is $U = cn$, $U \cdot U = -c^2$. Proper time is in seconds with a declared absolute epoch map. The mean-expansion redshift $\bar{z} = a^{-1} - 1$ is distinct from a ray's observed redshift. The selected interval is $\bar{z} = 20$ to 4. The locked conditional IC supplies no REC donor four-velocity or nuclear dataset. Its time origin is each matched $\bar{z} = 20$ slice; an absolute cosmic age is unspecified. Plot axes must therefore read "Elapsed proper time since $\bar{z} = 20$.

2 Locked conditional scenario and continuous source

The rounded reference cosmology is

$$H_{0,\mathrm{FLRW}} = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_m = 0.315,$$
$$\Omega_b = 0.0224/0.674^2, \qquad \Omega_r = 9.2 \times 10^{-5},$$
$$\Omega_\Lambda = 1 - \Omega_m - \Omega_r.$$

This is a Planck-like reference choice, not a joint likelihood draw. The selected helium mass fraction is 0.2454; $T_{\mathrm{CMB},0} = 2.7255$ K. The signed family uses $q_{\mathrm{amp}} = 0.01$ at $a_* = 1/21$.

The gas starts at $x_{\rm HII} = 2 \times 10^{-4}$, $x_{\rm HeII} = x_{\rm HeIII} = 0$, with the explicit approximate cooling IC

$$T_* = \frac{2.7255\, 21^2}{150(0.0224/0.023)^{0.4}}\,\mathrm{K}. \tag{2}$$

These are conditional freeze-out approximations, not exact recombination output. The archival 6.137 K is not a matching donor; 20 K is a separately named early-heating sensitivity. Initial astrophysical ionizing photons are explicitly zero in the selected source-onset IC.

The continuous source uses the selected Madau–Dickinson analytic law (equations 15 and 22, Figure 16 of the source listed below),

$$\psi(z) = \frac{0.015(1+z)^{2.7}}{1 + [(1+z)/2.9]^{5.6}}, \tag{3}$$

$$\log_{10} I_{\rm ion} = \frac{1}{0.00038 Z_*^{0.227} + 0.01858} - \log_{10}(2.55), \qquad Z_* = 0.002. \tag{4}$$

Here ψ has units $M_\odot\,\mathrm{yr}^{-1}\,\mathrm{cMpc}^{-3}$ and $I_{\rm ion}$ has photons/s per $(M_\odot/\mathrm{yr})$, so $\dot{n}_{\gamma,c} = f_{\rm esc} I_{\rm ion} \psi$ needs no extra year conversion. Baseline $f_{\rm esc} = 0.2$; a named sensitivity uses 0.25. The use above $z = 8$ is declared phenomenological continuation, not measured high-redshift calibration.

The photon-number spectrum is proportional to $E^2/[\exp(E/k_B T_*) - 1]$ with stellar $T_* = 50000$ K, normalized only on 13.6–100 eV. Its O-star-like blackbody is an idealized conditional spectrum. The implementation splits at $13.6, 24.59, 54.42, 100$ eV, with 16 log cells per segment, photon-number weights and cell mean energies. A preserved first quadrature failure led to the authorized Simpson subpanel increase from 8 to 32; independent moment checks then passed at 10^{-7}. Gauss–Legendre μ weights use 8 nodes, refined to 16/32, and one axisymmetric azimuth representative. A fixed covariant-ray driver must map local isotropic injection using its solid-angle Jacobian and normalize once. Source rates already integrate over 4π.

Proper injection is $\dot{n}_{\gamma,c}/(\mathrm{Mpc}_{\rm cm}^3 a^3)$ and per-H injection is $\dot{n}_{\gamma,c}/(\mathrm{Mpc}_{\rm cm}^3 n_{\rm H0})$. The module creates continuous photons, not imposed UVB photo rates or heat. The omitted blackbody tail is a separate 100–500 eV diagnostic; it is not injected outside support. The selected model still requires evolved histories and quantified omitted radiation-stress/CMB approximation diagnostics before admission.

3 Atomic gas and energy ownership

The evolved fractions are $(x_{\rm HII}, x_{\rm HeII}, x_{\rm HeIII})$, with neutral fractions supplied by element complements. The electron census is

$$n_e = n_{\rm H}[x_{\rm HII} + f_{\rm He}(x_{\rm HeII} + 2x_{\rm HeIII})], \qquad f_{\rm He} = n_{\rm He}/n_{\rm H}. \tag{5}$$

All densities are proper cm^{-3}. The ideal monatomic EOS is $u = \frac{3}{2} k_B T (n_{\rm H} + n_{\rm He} + n_e)$ and the evolved thermal variable is $w = u/n_{\rm H}$ in erg/H.

The selected point provider binds existing Grackle 3.4.1 functions at commit `af793949` (full pinned identity in the source manifest) through the repository's IGM implementation[2]. Its operational domain is $1 \le T/\mathrm{K} \le 10^6$; this range is numerical support, not measured empirical fit accuracy. Case A capture escape, clumping one, and primary-only photoelectron heating are selected. Source CI floors and CE caps remain diagnostic outputs. DR chemistry and owned cooling are off at $T \le 9284$ K, with excluded raw cooling retained diagnostically. CR, HH, RCT, metals, molecules, dust, secondary electrons and metastable CI are absent.

A common photo event owns the ionization, binding-energy increment, and excess thermal power. CI uses the same chemistry event and binding threshold as its thermal sink. RR/DR capture owns both released binding energy and kinetic cooling in escape. CE and free-free cooling add no chemistry events. CMB exchange is signed, with opposite gas/CMB reservoir rates. Expansion work on gas is $2Hu$. No recombination escape photon count is manufactured.

4 Radiation and matched source conventions

The isotropic proper-frame source and initial radiation are reflection-even. Energy and angle characteristics satisfy

$$\dot{E} = -E[H + s(3\mu^2 - 1)], \qquad \dot{\mu} = -3s\mu(1 - \mu^2). \tag{6}$$

The quadrupole is retained; dipole and axial force are tested only under the stated parity assumptions. Source history, energy support, proper/comoving conversion and absorption/work/boundary owners must be bound before numerical execution. Externally supplied point photo rates alone are not a source history.

5 Abundance normalization and mandatory products

For the selected H/He-only model $n_b = n_{\mathrm{H}} + 4n_{\mathrm{He}}$. Report $Y_i = n_i/n_b$ for every charge state, and $y_e = n_e/n_b$ separately for electrons. Element fractions and baryon-number shares A_iY_i are distinct quantities. The charge-state baryon sum is one only for the complete selected inventory; explicitly omitted reservoirs would permit a tracked sum below one. Donor nuclear abundance does not equal evolved ionic abundance. D, T, ^{3}He and Li charge states remain unimplemented optional extensions.

Mandatory products are abundance versus proper time, time-ordered parametric abundance versus T_{gas}, matched FLRW/Bianchi comparisons, and signed response versus numerical resolution. A nonmonotone temperature trajectory is never sorted into an apparent single-valued curve. Hydrogen 0.1/0.5/0.9 crossings are measured from saved histories; noncrossing is censored or not reached. Homogeneous x_{HII} is not Q_V.

6 Numerical validation and results intake

Point implicit residuals must use the same provider/model/domain as accepted steps. Positivity, nuclei/charge conservation, photon and energy ledgers, rejected trials, raw failures and identified checkpoints are required. A baseline and two refinement levels provide empirical resolution evidence; no FT03 interval certificate or rigorous continuum proof is inherited. One identified source variant and one IC variant quantify sensitivity without interpreting it as posterior uncertainty.

Product	Current intake status
Provider binding and original-C parity	See AX02 implementation receipt
Full FLRW and signed-shear histories	Full interval pending; partial FLRW below
Time/temperature abundance figures	Actual short-pilot diagnostics below
Convergence and source/IC sensitivity	Not yet inserted
Independent scoped scientific decision	Pilot-only PASS; full campaign pending

The retained F08 endpoint comparisons have their original synthetic prescribed-discrete scope. Their 15 endpoint rows are not reconstructed time histories and do not replace these campaign products. A resolved null or supported upper bound is an acceptable result; detection is not required. Segment Thomson depth may be reported as a secondary diagnostic; an unknown observer tail is neither zero nor a barrier to the abundance-history goal.

7 Completed pilot abundances and diagnostic figures

The revised output-cadence run completed the entire selected pilot interval $\bar{z} = 20$ to 19.8 for $q_{\rm ref} = 0$ and $+0.01$. Each case recorded 3882 accepted steps, 11700 actual tentative starts and 81924 RHS evaluations; each has 3883 saved scalar states including the initial state. The independent decision is `PASS_SCOPED_CONDITIONAL_PILOT_EXECUTION`. It checks the represented short-interval normalization, conservation, source-bound numerical controls and sampled perturbative feedback diagnostics. It does not admit the full campaign or a converged shear response.

Pilot quantity	FLRW	Positive shear
Endpoint $\bar{z}$	19.8	19.8
Elapsed proper time (Myr)	2.580698929	2.580571740
Endpoint $T_{\rm gas}$ (K)	22.565863657	22.565143160
Endpoint $x_{\rm HII}$	$5.243353784 \times 10^{-4}$	$5.243193221 \times 10^{-4}$
Measured wall time (s)	87.721678	108.308765
Measured CPU time (s)	87.573524	108.263777
GNU time peak RSS (KiB)	3800	4076

The figures use the saved nuclear charge-state number abundance $Y_i = n_i/(n_H + 4n_{\rm He})$. Electrons use $y_e = n_e/(n_H + 4n_{\rm He})$, are labelled charge census, and are excluded from nuclear baryon inventory. The source script checks the saved normalization against proper densities without changing input values. Initially exact zero He II and He III are masked only on the logarithmic display. All positive saved values are inside the plotted range; no fabricated floor, state clipping or temperature sorting is used. Very small plotted values do not carry an empirical fit-accuracy guarantee. Component and pilot residual checks do not establish the accuracy of tiny trace abundances. Displaying values near 10^{-34} does not imply 34 digit precision; global time, energy and angular convergence has not yet been measured.

The endpoint ionic and temperature differences remain below the frozen absolute response floors 10^{-7} in fraction and 0.01 K in temperature. No empirical refinement estimate is available yet, so this report issues neither a resolved shear detection nor a resolved null or validated upper bound.

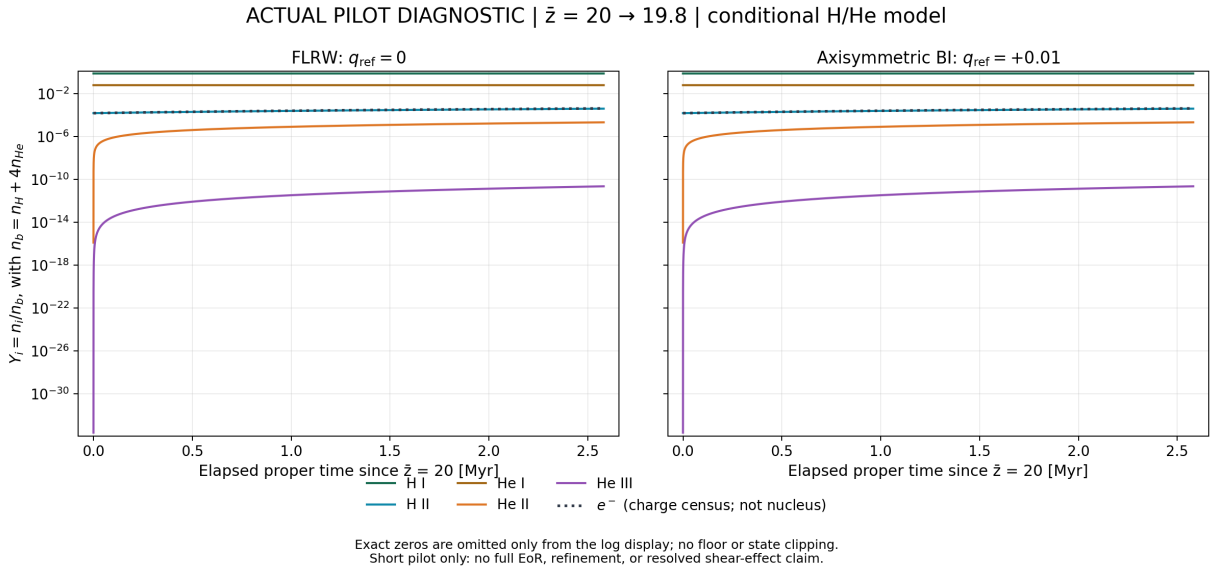


Figure 1: Actual conditional pilot abundance histories on the short $\bar{z} = 20$ to 19.8 interval. The clock is elapsed proper time since the selected initial slice, in Myr, not inferred absolute cosmic age. H/He charge states and the separately labelled electron census use the complete selected H/He baryon denominator. Exact zeros are omitted only on the log display; no state floor or clipping is applied. Tiny trace values are displayed as saved, without a claim of validated trace accuracy or numerical precision. Component and pilot residual checks do not establish global convergence. This is a diagnostic pilot product, not a full EoR, convergence or resolved-response figure.

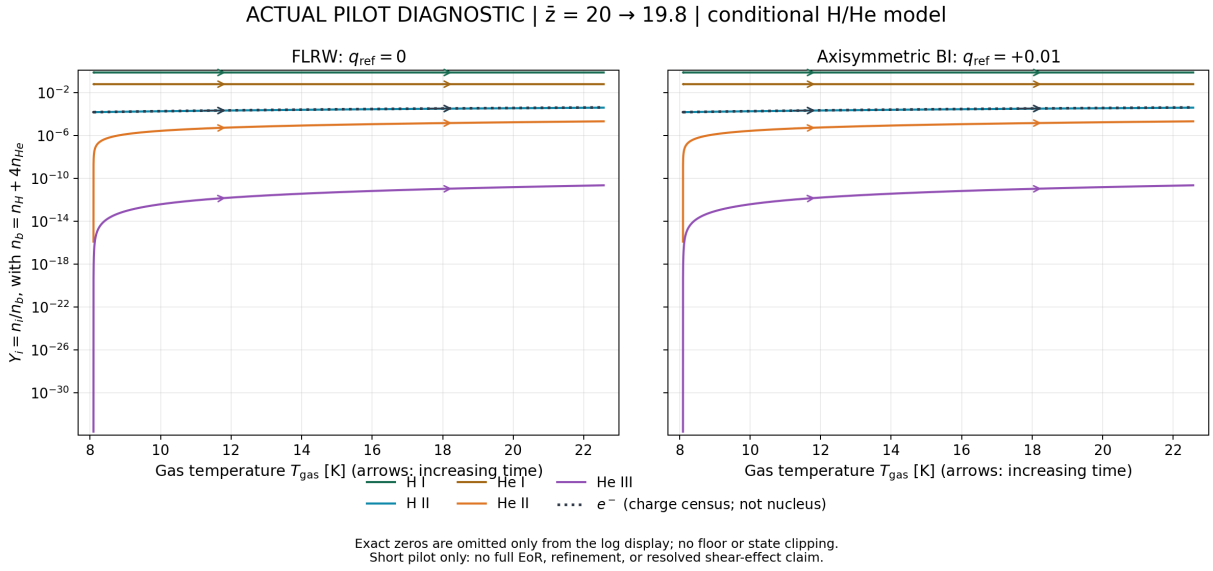


Figure 2: The same actual pilot abundances plotted parametrically against gas temperature in saved time order. Arrows indicate increasing time; the curve is never sorted by temperature, including the tiny early cooling excursion. Exact-zero, normalization and scope conventions match the preceding figure. These short pilot curves do not replace the required full-history BBN-style products.

8 Observed partial FLRW pilot

The first actual FLRW pilot used the reviewed binary identity `4080898d7770` and attempted the selected $\bar{z} = 20$ to 19.8 interval. It stopped checkpointed at $\bar{z} = 19.86586760536834$ because the exact prewrite aggregate output cap rejected the next output batch. The elapsed proper time at that partial checkpoint was $5.440334630231574 \times 10^{13}$ s. This is a represented partial coupled history; the requested pilot endpoint was not reached, and the positive-shear pilot was not started.

Measured pilot quantity	Value
Accepted steps	2870
Actual tentative steps	8667
Rejected adaptive candidates	18
RHS evaluations	60693
Measured wall time (s)	64.233055
Measured CPU time (s)	64.128211
GNU time peak RSS (KiB)	3724
Maximum recorded Newton residual	2.712×10^{-13}
Last recorded local estimator	3.817×10^{-6}

The exact terminal error was `AXBI_OUTPUT_CAP`; no numerical failure was reported as the stop reason. The source/model and tolerances were unchanged. Original history, abundances, spectra, ledgers, raw streams, source/config identities and checkpoint remain preserved. A lossless archive contains all 66 original regular files, 234862901 bytes before compression and 26097828 bytes compressed; every archived file was checked against its input size and SHA. This packaging verification does not admit the trajectory scientifically. Full-interval results, paired response and convergence remain pending.

The authority selected one output-cadence repair: retain every accepted scalar abundance, history, ledger and feedback row, and save full unaltered spectra initially, at the existing 60-step or 30-second checkpoints, and terminally. The original failed attempt is retained. A new explicitly revised allocation and changed-output review must bind any subsequent pilot binary and source identity; no automatic retry was performed.

9 Checkpointed ongoing full-stage progress

The immutable snapshot captured at 2026-10-14 08:33:57 UTC uses the existing source-bound binary `32d11e733155`. Both cases remain incomplete relative to the selected $\bar{z} = 4$ endpoint. Each plotted prefix ends exactly at its separately captured durable checkpoint.

Conditional saved quantity	ZERO	Positive shear
Checkpoint $\bar{z}$	12.7867563	14.0201997
Elapsed proper time (Myr)	157.136676	116.631221
T_{gas} (K)	1285.87339	862.424607
x_{HII}	0.037996897	0.0242490861
x_{HeII}	0.0399903547	0.0254391198
x_{HeIII}	4.47551775e-06	1.83323659e-06

These rows have different mean-redshift and elapsed-time endpoints. Their apparent differences are not a matched shear response. The figures show actual chronological partial abundance histories under the fixed conditional model; the selected full interval, global convergence and resolved response have not been established. Tiny trace values are displayed as saved and carry no validated precision claim. Electrons remain a separately labelled charge census.

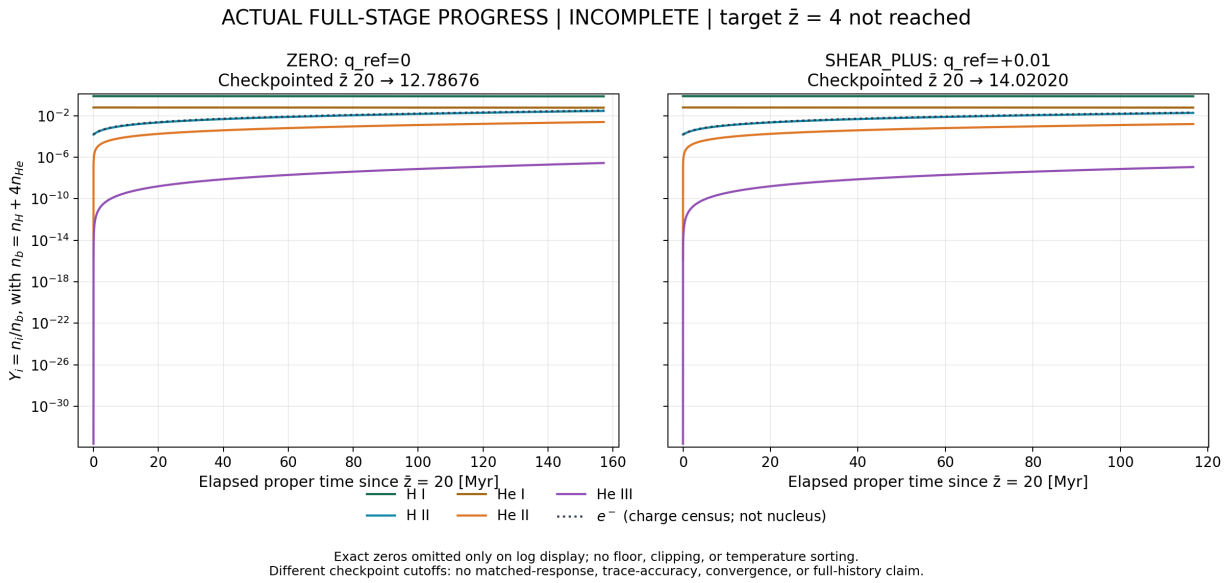


Figure 3: Checkpoint-bounded actual elapsed proper-time abundance progress. The two cases have different represented endpoints; target mean redshift4 is not reached. This is incomplete conditional execution evidence, not a matched response or full-history result.

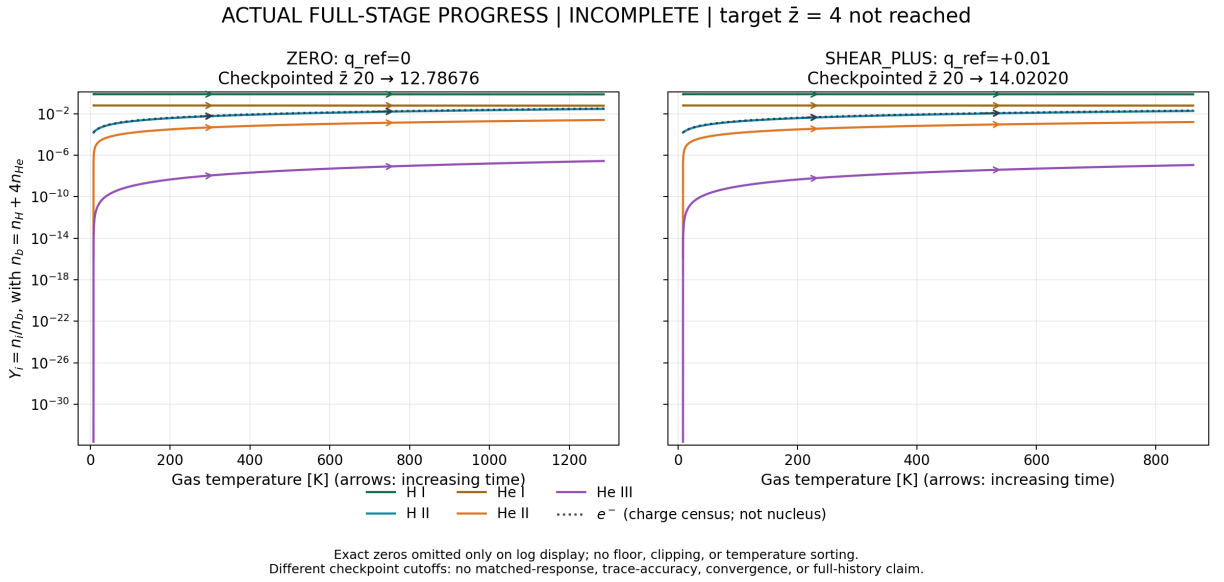


Figure 4: The same partial abundance trajectories against gas temperature in saved time order, with arrows for increasing time. Temperature is not sorted. Exact zeros are masked only in log display, and positive states are neither floored nor clipped. No trace-accuracy or convergence claim is inferred.

10 Reproduction

Future release instructions must identify actual driver/config/source versions, complete case argv, cumulative budgets, raw stdout/stderr, checkpoints and output CSV/figure scripts. Current reproduction evidence includes both completed pilot histories, their raw streams, checkpoints and frozen identities, and the retained failed attempt. The source manifest and figure manifest bind the actual products. The plotting script is `plot_pilot_abundances.py`; run it from its figure directory or repository root. Compile this manuscript from the repository root so its figure paths resolve. Full-campaign reproduction remains pending. Publication and a compilable manuscript do not constitute scientific completion.

References

[1] P. Madau and M. Dickinson, Cosmic Star-Formation History, equations 15 and 22 and Figure 16, `https://arxiv.org/html/1403.0007`.

[2] Grackle development team, Grackle 3.4.1 pinned rate functions, `https://github.com/grackle-project/grackle/blob/af7939494ce65007887ada7b98d1813df6843346/src/clib/rate_functions.c`. Repository literal-C fixture and retained license provide the independent point reference.